

Qatar-1b

R-2695

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160622 · created 2026-08-31T23:43:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457561.82459 +/- 0.00075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.014
Transit depth (Rp/Rs)^2	2.84 +/- 0.48 [%]
Orbital Inclination (inc)	86.0 +/- 1.33
Airmass coefficient 1 (a1)	0.876 +/- 0.019
Airmass coefficient 2 (a2)	0.105 +/- 0.037
Scatter in the residuals of the lightcurve fit is	2.13 %
Best Comparison Star	#3 - [565, 151]
Optimal Method	PSF photometry
Transit Duration (day)	0.0785 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.169 ± 0.014 vs 0.1463 (deviation 1.32σ)
Duration fit vs published	0.0785 d vs 0.0692 d (deviation 1.69σ)
Mid-time O-C	-2.0 min
Depth SNR	9.8
Residual scatter	2.13 %

Light curve

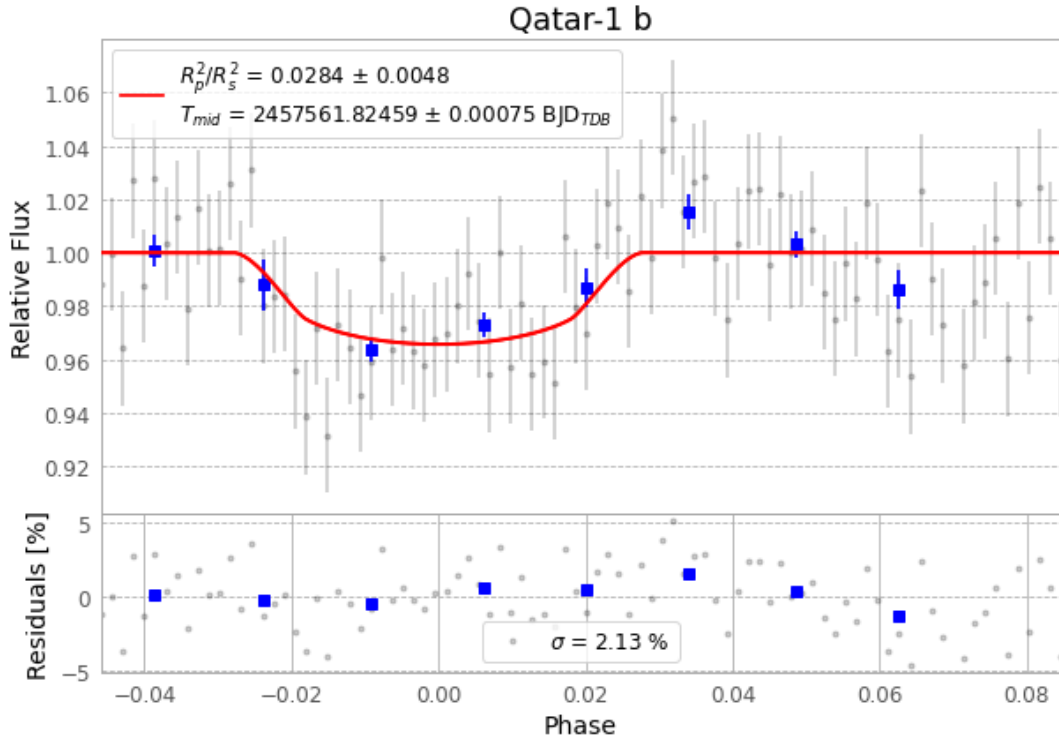


Figure 1 — FinalLightCurve_Qatar-1 b_2016-06-21.png (SHA-256 64a280bd997bd52e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 98fb5f181cafca71...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

90 FITS frames (59 MB), Qatar-1160622061212.FITS → Qatar-1160622103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160622.log (ecab02f90bc4...), FinalLightCurve_Qatar-1 b_2016-06-21.png (64a280bd997b...), FinalLightCurve_Qatar-1 b_2016-06-21.csv (27fe65e944f5...)

Compute resources

Wall time 149.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 23:43Z.